MORADABAD INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SESSIONAL TEST 1

SET-1

Course: B.Tech. **Semester: 5th** **Session: 2024-25**

Section: A/B/C **Subject: MLT (DE-II)** **Sub. Code: BCS-055**

Max. Marks: 20 **Time: 01 Hr.** **Date: 06-11-2024**

Q. No. :	1	2	3	4	5	6
CO No.:	1	1	2	1	2	2
Bloom's Level	K1	K2	K4	K2	K1	K1 / K3

CO1. To understand the need for machine learning for various problem solving.

CO2. To understand a wide variety of learning algorithms and how to evaluate models generated from data

Note:

1) ***This paper contains two sections. Section (A) & (B).***
2) ***All sections are compulsory.***

SECTION (A)

Attempt all questions. Each question carries 2 marks.

1. Differentiate Data Science and Machine Learning, also write at least 3 applications of supervised Learning. [2 Marks]
2. Explain the different approaches of machine learning. [3 Marks]
3. Draw the Final Design of Checkers Learner Problem. [3 Marks]

(OR)

Discuss the Bayesian Classification and why we need it?

SECTION (B)

Attempt all questions. Each question carries 3 marks.

4. Explain the steps involve in knowledge Discovery in Database process with suitable diagram. [4 Marks]

(OR)

What do you understand by well posed learning, explain with the automated vehicle example.

5. Explain Linear Regression in detail, also differentiate it with Logistic Regression. [4 Marks]
6. Explain all the steps involve to solve any Machine Learning Problem, [4 Marks]

(OR)

The values of independent variable x and dependent value y are given below:

X	0	1	2	3	4
Y	2	3	5	4	6

Design the least square regression line model y= ax + b. Estimate the value of y when x is 14.

******All the Best******

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST 2
SET-1

Course: B.Tech. **Semester: 5th** **Session: 2024-25**
Section: A/B/C **Subject: MLT (DE-II)** **Sub. Code: BCS-055**
Max. Marks: 20 **Time: 01 Hr.** **Date: 21-12-2024**

Q. No. :	1	2	3	4	5	6
CO No.:	3	4	5	3	4	5
Bloom's Level	K2	K2	K4	K2	K1	K1 / K3

CO3. To understand the latest trends in machine learning.
CO4. To design appropriate machine learning algorithms and apply the algorithms to a real world problem.
CO5. To optimize the models learned and report on the expected accuracy that can be achieved by applying the models.

Note:

1) ***This paper contains two sections. Section (A) & (B).***
2) ***All sections are compulsory.***

SECTION (A)

Attempt all questions. Each question carries 2 marks.

1. Differentiate (a) Overfitting and Underfitting, (b) Pre-pruning and Post-pruning [2 Marks]
2. How to deal with the overfitting problem in backpropagation learning, Explain? [3 Marks]
3. Explain the Off policy used in RL algorithm also explain the ON policy temporal difference learning method. [3 Marks]

SECTION (B)

Attempt all questions. Each question carries 3 marks.

4. Differentiate Attribute Selection Measures ID3 and C4.5. also calculate the information gain of student from the following data set. [4 Marks]

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

5. Describe the steps involved in the training of CNN Model in detail. [4 Marks]
6. Explain the work flow of Genetic Algorithm in detail with its application, also explain the terms used in PCA algorithm. [4 Marks]

******All the Best******